SABER: Support-Conditioned Adaptive Bellman Error Reweighting for Sparse HER Actor-Critic

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Abstract

Hindsight Experience Replay (HER) converts failed experience into goal-conditioned training data through achieved-goal relabeling. Most HER extensions alter goal selection, replay selection, or target construction. We instead ask whether per-sample squared Bellman residuals should remain equally weighted once relabeling and replay sampling are fixed. We introduce SABER (Support-Conditioned Adaptive Bellman Error Reweighting), a critic-only objective. SABER upweights a sample's squared Bellman residual when a sufficiently populated local replay set from the same discretized state–goal–horizon context contains an action in the current-policy action bin. It normalizes the weights to unit minibatch mean and uses the current-policy action only for the support query while retaining Bellman regression at the replay action. HER relabeling, uniform replay sampling, the actor objective, Bellman targets, network architectures, and the update-to-data (UTD) ratio remain unchanged. Under matched UTD-1 training on FetchPickAndPlace, SABER raises mean normalized success AUC from 0.539 to 0.609 with DDPG, from 0.372 to 0.442 with TD3, and from 0.471 to 0.583 with SAC. Paired-bootstrap 95% confidence intervals for the AUC differences exclude zero for TD3 and SAC but include zero for DDPG. On DrawerOpen with DDPG, SABER raises mean AUC from 0.207 to 0.754 and mean final success from 0.390 to 1.000. Controlled comparisons, ablations, and diagnostics examine alternative allocation mechanisms and characterize SABER's support signal. These gains, achieved under the same one-update budget and an otherwise unchanged HER pipeline, establish post-HER critic-loss allocation as a distinct design axis capable of substantial performance improvements.

1 Introduction

A central challenge in sparse-reward goal-conditioned reinforcement learning is obtaining useful supervision when successful trajectories are rare. Hindsight Experience Replay (HER) addresses this challenge by relabeling failed experience with achieved goals (Andrychowicz et al. 2017). Once sampling and relabeling are fixed, however, standard HER actor–critics assign equal coefficients to all sampled Bellman residuals. This leaves a separate question: should critic supervision remain uniform when the critic regresses at replay actions but the actor queries the critic at current-policy actions?

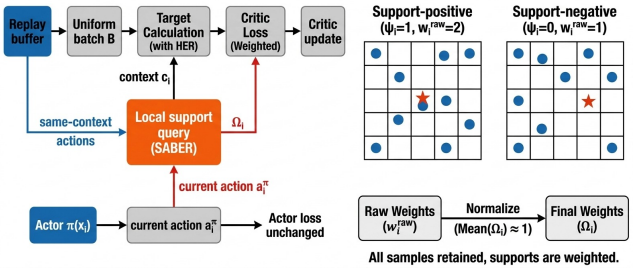


Figure 1: SABER intervenes after HER relabeling and before critic-loss aggregation. Stage 3 uses the current-policy action only to query local support; Stage 4 keeps Bellman regression at the replay action.

We introduce SABER (Support-Conditioned Adaptive Bellman Error Reweighting), a critic-only post-HER objective. SABER tests whether replay actions from the same discretized state–goal–horizon context occupy the current-policy action bin, upweights support-positive residuals, and normalizes weights to unit minibatch mean. The current-policy action is used only for this support query; regression remains at the replay action. Relabeling, replay sampling, rewards, Bellman targets, actor optimization, architectures, and the update budget remain unchanged.

Our contributions are: (i) identifying post-HER critic-loss allocation as distinct from goal selection, replay prioritization, and target construction; (ii) introducing a normalized local-support objective that preserves the underlying HER actor–critic pipeline; and (iii) evaluating its cross-backbone repeatability, query and count-matched controls, update-budget trade-offs, diagnostics, and task-dependent success and failure boundaries.

2 Related Work

Goal-conditioned methods modify several stages around HER. HGG and CHER generate or schedule hindsight goals, EBP prioritizes energetic episodes, and HGR prioritizes both episodes and transition–future-goal pairs by TD error with two-stage importance correction (Ren et al. 2019;

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Fang et al. 2019; Zhao and Tresp 2018; Luu and Yoo 2021). MHER changes relabeled returns, ARCHER rescales hindsight rewards, and USHER corrects hindsight-induced value bias (Yang et al. 2021; Lanka and Wu 2018; Schramm et al. 2023). These methods change goal construction, experience selection, rewards, or targets. In contrast, SABER intervenes only after goal construction, replay sampling, rewards, and Bellman targets have been fixed: it preserves the relabeled minibatch and changes only the coefficient assigned to each critic Bellman residual.

General replay methods redistribute experience independently of HER. PER samples by TD error with importance correction, AER selects state-similar transitions, and ReF-ER suppresses far-policy samples while regularizing the actor (Schaul et al. 2015; **?**; Novati and Koumoutsakos 2019). SABER instead preserves minibatch membership and uses conditional action support only to set critic-loss coefficients.

Closer critic-reweighting methods learn different signals: likelihood-free importance weighting estimates a policy-to-replay density ratio, DisCor estimates target accuracy, and ReenGAGE adds goal-gradient distillation (**???**). SABER learns neither a density ratio nor an error model and adds no auxiliary target. Its distinction is a non-learned, sample-aligned current-policy action-support predicate applied to post-HER critic residuals.

3 SABER

3.1 Problem Formulation and the Post-HER Critic-Loss Allocation Gap

We consider a finite-horizon goal-conditioned Markov decision process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{G}, P, r, \gamma, H)$. At time t, the environment is in state $s_t \in \mathcal{S}$, the agent receives observation $o_t = \omega(s_t)$, and the desired goal is $g \in \mathcal{G}$. The goal-conditioned input is $x_t = (o_t, g)$, and the policy and critic are denoted by $\pi_\theta(a \mid x)$ and $Q_{\phi_k}(x, a)$, respectively. Let $m : \mathcal{S} \to \mathcal{G}$ map a state to its achieved goal. The environment provides the sparse goal-conditioned reward

$$r(s_t, a_t, s_{t+1}, g) = \begin{cases} 0, & d(m(s_{t+1}), g) \le \epsilon_g, \\ -1, & \text{otherwise.} \end{cases} \tag{1}$$

When successful trajectories are rare, replay collected under the original desired goal contains little direct successful supervision.

Future-goal HER makes such experience trainable by relabeling a transition at time t with an achieved goal from a later state:

$$\begin{aligned} &\tilde{g}_i = m(s_{t'}), && t' \in \{t+1, \dots, T\}, \\ &\tilde{r}_i = r(s_t, a_t, s_{t+1}, \tilde{g}_i). \end{aligned} \tag{2}$$

For a transition not selected for relabeling, $\tilde{g}_i$ remains the original desired goal. The resulting network inputs are $x_i = (o_i, \tilde{g}_i)$ and $x'_i = (o'_i, \tilde{g}_i)$. For backbone A, let y_i^{A} denote its standard Bellman target after relabeling and reward recomputation. For critic k, the replay-action Bellman residual is

$$\delta_{ik} = Q_{\phi_k}(x_i, a_i^r) - y_i^{\mathsf{A}}, \tag{3}$$

where a_i^r is the action stored in replay. Given a uniformly sampled and relabeled minibatch B, the standard HER critic objective is

$$\mathcal{L}_Q^{\mathrm{HER}} = \frac{1}{|B|} \sum_{i \in B} \sum_{k=1}^{n_Q} \delta_{ik}^2. \tag{4}$$

Thus, once replay sampling and relabeling are fixed, every sampled residual receives the same explicit coefficient.

HER determines which relabeled goals and rewards are used for learning, but it does not determine how critic fitting effort should be distributed among the residuals in the resulting minibatch. This gap may matter in off-policy actor–critic learning: the critic regresses Bellman targets at replay actions a_i^r, whereas the actor improves the policy by querying the critic at current-policy actions a_i^π. The amount of replay evidence near these policy queries can vary across state–goal contexts. We therefore ask whether observable local replay support for a_i^π can be used to allocate critic-loss weight while holding the relabeled minibatch and Bellman targets fixed.

Because this question requires both replay evidence and a replay-based critic objective, SABER is designed for replay-based off-policy actor–critic learning. We instantiate it with three representative continuous-control backbones: DDPG, which uses a deterministic actor and one critic; TD3, which uses a deterministic actor, twin critics, target-policy smoothing, and delayed actor updates; and SAC, which uses a stochastic entropy-regularized actor and twin critics (Lillicrap et al. 2015; Fujimoto, van Hoof, and Meger 2018; Haarnoja et al. 2018). These algorithms provide representative deterministic and stochastic off-policy structures while retaining the replay-based critic update required by SABER. SABER leaves each backbone's Bellman target, bootstrap convention, and actor objective unchanged.

3.2 SABER Overview and Intervention Point

SABER is not an additional neural-network layer or a replay sampler. It is a per-sample critic-loss reweighting objective inserted after uniform replay sampling, HER relabeling, reward recomputation, and Bellman-target construction, but before the per-sample critic losses are reduced to a minibatch mean. As illustrated in Fig. 1, SABER forms a local state–goal–horizon replay context for each relabeled sample and asks whether the corresponding current-policy action is represented in that context's replay-action support. It converts this binary query into a normalized critic weight. The current-policy action is used only to query support; the weighted Bellman regression remains evaluated at the original replay action.

This intervention keeps every sampled transition in the update and does not alter which goal, reward, or Bellman target is attached to that transition. SABER therefore isolates a post-HER allocation decision: given an already sampled and relabeled minibatch, which squared Bellman residuals should receive greater relative fitting pressure?

3.3 Current-Policy Local Support

For sample i, SABER constructs a task-specific discrete full support-context key

$$c_i = C(o_i, \tilde{g}_i, h_i), \tag{5}$$

where h_i is the remaining horizon. In the manipulation domains evaluated in Sec. 4, the full key summarizes task progression, agent–object interaction, goal-relative geometry, gripper state, and remaining horizon. Concretely, FetchPickAndPlace uses manipulation phase, hand–object distance, object–goal distance, object height, gripper gap, and remaining horizon. The Drawer tasks use interaction phase, hand–handle distance, handle–goal distance, handle position along the opening axis, gripper state, and remaining horizon. These features are discretized using fixed, task-specific bins.

To bound lookup cost, SABER first uses a lower-dimensional *coarse retrieval key* $C^{\mathrm{ret}}(o, h)$, constructed from a subset of the full-key components, only to retrieve candidate replay indices. It takes at most the N_{scan} most recent indices sharing the query's coarse retrieval key:

$$S_i = \mathrm{Tail}_{N_{\mathrm{scan}}}\left(\left\{j \in \mathcal{R} : C^{\mathrm{ret}}(o_j, h_j) = C^{\mathrm{ret}}(o_i, h_i)\right\}\right). \tag{6}$$

The coarse retrieval key is an indexing device, not the support criterion. Candidates are filtered using the full support-context key:

$$I_i = \{j \in S_i : C(o_j, \tilde{g}_i, h_j) = C(o_i, \tilde{g}_i, h_i)\}. \tag{7}$$

This equality is exact component-wise matching: a candidate is retained only when every discretized component of its full key equals the corresponding query component. The relabeled query goal $\tilde{g}_i$ is used to compute both sides of this comparison, so a candidate is not matched using its originally stored desired goal.

Each action dimension is normalized by its environment bounds and divided into K bins. Let $b_K(a)$ denote the resulting multidimensional action bin and

$$\mathcal{U}_i = \left\{b_K(a_j^r) \mid j \in I_i\right\} \tag{8}$$

the set of replay-action bins occupied in the local context. With $n_i = |I_i|$, SABER defines the support indicator

$$\begin{aligned}\psi_i = \mathbf{1}\{n_i \geq N_{\mathrm{local}}\}\,\mathbf{1}\{\min(n_i, M_{\mathrm{cap}}) \geq M_{\mathrm{avail}}\} \\ \cdot\,\mathbf{1}\{b_K(a_i^\pi) \in \mathcal{U}_i\}.\end{aligned} \tag{9}$$

The first two conditions require a sufficiently populated local context and sufficient capped support availability; the final condition tests whether the current-policy action bin has been observed in that context. The main setting uses

$$\begin{aligned}K = 5, \quad N_{\mathrm{scan}} = 512, \quad N_{\mathrm{local}} = 4, \\ M_{\mathrm{cap}} = 16, \quad M_{\mathrm{avail}} = 4.\end{aligned} \tag{10}$$

DDPG and TD3 use their deterministic actor action as a_i^π. For the support query, SAC uses the deterministic squashed mean of its actor while retaining stochastic actions in its standard actor and target objectives. The indicator uses neither Q-values nor TD errors. It is a binary occupancy test, and the current-policy action is not substituted for a_i^r in the Bellman residual.

3.4 Normalized Critic-Only Objective and Backbone Integration

SABER assigns the raw per-sample weight

$$w_i^{\mathrm{raw}} = 1 + \lambda_S \psi_i \tag{11}$$

and normalizes the weights by their minibatch mean:

$$\bar{w}_B = \frac{1}{|B|}\sum_{j \in B} w_j^{\mathrm{raw}}, \qquad \Omega_i = \frac{w_i^{\mathrm{raw}}}{\max(\epsilon_w, \bar{w}_B)}. \tag{12}$$

The main method uses $\lambda_S = 1$, so support-positive and support-negative samples receive raw weights two and one, respectively. If $p_B = |B|^{-1}\sum_{i \in B}\psi_i$ is the support-positive fraction, then, apart from the inactive numerical safeguard,

$$\Omega_i = \begin{cases} \dfrac{1+\lambda_S}{1+\lambda_S p_B}, & \psi_i = 1, \\ \dfrac{1}{1+\lambda_S p_B}, & \psi_i = 0. \end{cases} \tag{13}$$

This makes the allocation effect explicit: positive samples receive greater relative weight, negative samples remain in the update with positive weight, and the minibatch mean coefficient is one. If all samples share the same indicator value, normalization returns $\Omega_i = 1$ for every sample.

The SABER critic objective is

$$\mathcal{L}_Q^{\mathrm{SABER}} = \frac{1}{|B|}\sum_{i \in B}\Omega_i \sum_{k=1}^{n_Q}\delta_{ik}^2. \tag{14}$$

We use $n_Q = 1$ for DDPG and $n_Q = 2$ for TD3 and SAC; for twin-critic backbones, the same Ω_i is applied to both squared residuals. SABER therefore changes the relative coefficients of existing residual-gradient terms without adding samples, targets, or auxiliary losses. Mean-one normalization controls the average coefficient but does not assert that the numerical loss or gradient norm is unchanged, because weights may correlate with residual magnitude.

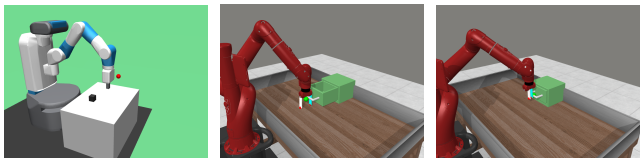
The main variant is critic-only:

$$\mathcal{L}_\pi^{\mathrm{SABER}} = \mathcal{L}_\pi^{\mathsf{A}}. \tag{15}$$

Critic-only weighting is a design choice, not a limitation that makes actor weighting impossible. We select it because it tests post-HER Bellman-loss allocation most directly while minimizing changes to other learning components. It preserves the DDPG policy objective, TD3 delayed-policy rule, and SAC entropy-regularized actor objective, thereby making an observed effect easier to attribute to critic fitting. We do not claim that critic-only weighting is always the performance-optimal scope.

One SABER update therefore proceeds as follows: sample replay transitions uniformly; apply future-goal HER and recompute rewards; construct the backbone's unchanged Bellman targets; query current-policy local support; compute ψ_i, w_i^{raw}, and Ω_i; update the critic with $\mathcal{L}_Q^{\mathrm{SABER}}$; and perform the backbone's original actor and target-network updates. HER probability, replay sampling, network architecture, optimizer, exploration process, Bellman targets, and UTD remain unchanged.

Finally, support positivity is a binary, discretized occupancy proxy, not an estimate of action density, uncertainty, target accuracy, or target variance. SABER does not directly regress the critic at a_i^π, discover successful actions absent from replay, or change exploration. Its task-specific



(a) FetchPickAndPlace (b) DrawerOpen (c) DrawerClose

Figure 2: Illustrative simulator renderings of the experimental tasks. The panels are not evaluation rollouts.

keys and bounded retrieval can merge inappropriate transitions or fragment useful experience. These properties delimit the method's scope while preserving its central intervention: normalized allocation of critic fitting within a fixed, HER-relabeled minibatch.

4 Experiments

4.1 Common Experimental Protocol

Tasks, cohorts, and training protocol. We evaluate FetchPickAndPlace-v4 (FPP; horizon 50, 1M environment steps) with DDPG, TD3, and SAC, and Meta-World DrawerOpen and DrawerClose (horizon 200, 2M steps) with DDPG and TD3. Primary comparisons use paired seeds 1–5, future-goal HER probability 0.8, and UTD 1. We make no paired claims for SAC on the Drawer tasks because no complete matched HER–SABER cohort exists: the available DrawerOpen SAC+HER runs remain at zero success for all five seeds, and neither matched SABER runs nor DrawerClose SAC results were collected. Sparse rewards and future relabeling follow HER (Andrychowicz et al. 2017); backbone targets and policy updates follow DDPG, TD3, and SAC (Lillicrap et al. 2015; Fujimoto, van Hoof, and Meger 2018; Haarnoja et al. 2018). Shared continuous-control settings are fixed within each matched task–backbone cohort without method-specific retuning; complete values, implementation details, and provenance are provided in the supplement.

Evaluation and statistics. Policies are evaluated deterministically over 20 episodes at each checkpoint. Paired analyses use only exact shared evaluation steps, without smoothing, interpolation, or filling. Normalized success AUC is the trapezoidal area divided by the shared step span; final success and Hit90 (the first checkpoint with success ≥ 0.9, reported with reached-seed coverage) are secondary. We report seed means and sample standard deviations; paired claims use within-seed differences and 10,000-resample percentile-bootstrap 95% confidence intervals. Absolute AUC values are not compared across tasks with different horizons or budgets.

4.2 Primary Comparisons

Comparison setup. HER addresses sparse supervision through goal relabeling, but learning effort can be reallocated at several stages of the resulting training pipeline. We compare methods that intervene at three distinct points. HGR changes hindsight-experience selection by prioritizing episodes and transition–future-goal pairs according to TD error, with importance correction at both stages (Luu and Yoo 2021). PER changes replay selection by sampling replay transitions according to TD-error priorities and applying annealed importance correction (Schaul et al. 2015). SABER instead preserves both selections and changes only the critic-loss coefficients within the sampled and relabeled minibatch. These methods are therefore alternative interventions within the HER actor–critic pipeline rather than variants of a single mechanism.

We additionally include *TD-error critic-loss weighting* as a direct allocation control. It retains uniform replay and the sampled HER minibatch, computes detached TD-error priorities $p_i = (|\delta_i| + 10^{-6})^{0.6}$, caps their mean-normalized values at 2, and renormalizes the resulting critic coefficients to unit minibatch mean. Unlike PER, it uses no priority buffer, nonuniform replay, or importance correction. It therefore matches SABER's intervention point while replacing local replay support with TD error as the allocation signal. Together, the comparisons separate *where* learning effort is reallocated—hindsight selection, replay selection, or critic-loss allocation—from *which signal* drives that allocation. AER, high-replay-ratio HER, and REDQ-HER are retained as secondary scaling references because they additionally change HER usage, update budget, or architecture.

Table 1 summarizes final success, success AUC, and Hit90 across all task–backbone cohorts. Figure 3 complements these aggregate metrics by showing FPP learning dynamics across all three backbones.

Matched HER comparison. SABER achieves higher mean AUC than matched HER in all seven task–backbone cohorts while preserving replay sampling, HER relabeling, Bellman targets, actor optimization, architecture, and UTD. On FPP, mean AUC increases from 0.539 to 0.609 with DDPG, from 0.372 to 0.442 with TD3, and from 0.471 to 0.583 with SAC. Paired 95% confidence intervals exclude zero for TD3 and SAC but include zero for DDPG. The largest gain occurs on DrawerOpen DDPG, where AUC increases from 0.207 to 0.754, final success from 0.390 to 1.000, and Hit90 coverage from 2/5 to 5/5 seeds. Mean AUC also increases under both DDPG and TD3 on DrawerClose. Thus, reallocating critic loss alone can consistently improve the mean learning outcome of an otherwise fixed HER pipeline.

HGR comparison. HGR and SABER address different allocation decisions in HER-based learning: HGR changes which hindsight experiences enter an update, whereas SABER changes the relative critic-fitting contribution of experiences already selected. SABER achieves higher mean AUC on all three FPP backbones and on DrawerOpen DDPG. HGR is slightly higher on DrawerOpen TD3 and higher on both DrawerClose backbones. On DrawerClose, however, HGR's higher AUC does not imply stronger terminal stability: final success is 0.080 versus SABER's 0.280 with DDPG and 0.500 versus 0.590 with TD3. The comparison therefore indicates different task-dependent operating regimes rather than a universal ordering. In our recorded implementations, HGR has lower throughput than SABER in every evaluated cohort; these hardware-specific measurements are reported separately and are not used as a performance claim.

Table 1: Primary post-HER comparisons over paired seeds 1–5. Entries are mean ± sample standard deviation; Hit90 is the median first step reaching 90% success, with reached-seed coverage in parentheses. Repeated *HER+* prefixes are omitted for compactness. SABER denotes our method, and its reported values are shown in bold. FPP SAC uses the exact common 50k–1M evaluation grid.

FetchPickAndPlace (FPP; 1M)

Method	Final	AUC	Hit90
DDPG			
SABER	**0.960 ± .042**	**0.609 ± .038**	**540k (5/5)**
HER	0.940 ± .042	0.539 ± .125	600k (5/5)
HGR	0.540 ± .365	0.338 ± .149	930k (2/5)
TD-error weighting	0.930 ± .104	0.568 ± .069	630k (5/5)
PER	0.780 ± .438	0.460 ± .238	640k (4/5)
TD3			
SABER	**0.970 ± .045**	**0.442 ± .035**	**840k (5/5)**
HER	0.840 ± .188	0.372 ± .032	980k (3/5)
HGR	0.610 ± .198	0.367 ± .034	630k (1/5)
TD-error weighting	0.490 ± .171	0.351 ± .021	630k (1/5)
PER	0.660 ± .204	0.369 ± .034	1.00M (1/5)
SAC			
SABER	**0.990 ± .022**	**0.583 ± .030**	**750k (5/5)**
HER	0.790 ± .288	0.471 ± .079	800k (3/5)
HGR	0.760 ± .251	0.484 ± .071	800k (2/5)
TD-error weighting	0.990 ± .022	0.544 ± .068	800k (5/5)
PER	0.990 ± .022	0.690 ± .046	550k (5/5)

DrawerOpen (2M)

Method	Final	AUC	Hit90
DDPG			
SABER	**1.000 ± .000**	**0.754 ± .044**	**500k (5/5)**
HER	0.390 ± .534	0.207 ± .287	960k (2/5)
HGR	0.900 ± .224	0.688 ± .142	530k (5/5)
TD-error weighting	0.800 ± .447	0.715 ± .090	610k (5/5)
PER	0.990 ± .022	0.825 ± .044	330k (5/5)
TD3			
SABER	**0.590 ± .539**	**0.362 ± .338**	**780k (3/5)**
HER	0.190 ± .425	0.127 ± .283	740k (1/5)
HGR	0.580 ± .531	0.371 ± .343	760k (3/5)
TD-error weighting	0.400 ± .548	0.202 ± .284	1.015M (2/5)
PER	0.600 ± .548	0.406 ± .374	720k (3/5)

DrawerClose (2M)

Method	Final	AUC	Hit90
DDPG			
SABER	**0.280 ± .251**	**0.324 ± .082**	**20k (5/5)**
HER	0.160 ± .182	0.251 ± .030	70k (5/5)
HGR	0.080 ± .091	0.349 ± .184	100k (5/5)
TD-error weighting	0.400 ± .166	0.313 ± .098	130k (5/5)
PER	0.460 ± .363	0.308 ± .049	60k (5/5)
TD3			
SABER	**0.590 ± .539**	**0.549 ± .422**	**80k (5/5)**
HER	0.350 ± .480	0.329 ± .348	70k (5/5)
HGR	0.500 ± .457	0.602 ± .305	80k (5/5)
TD-error weighting	0.200 ± .447	0.369 ± .275	30k (5/5)
PER	0.600 ± .548	0.524 ± .379	120k (5/5)

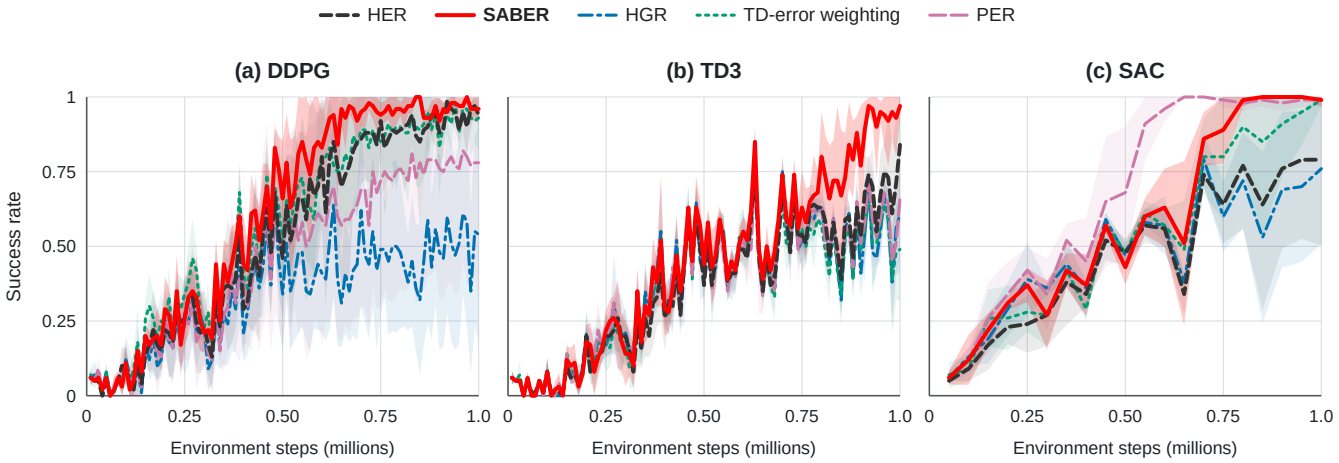


Figure 3: FetchPickAndPlace learning curves for the primary comparisons across DDPG, TD3, and SAC. Lines show mean deterministic evaluation success over paired seeds 1–5, and shaded regions show sample standard deviation. DDPG and TD3 use the exact 10k evaluation grid, while SAC uses the exact shared 50k grid. No smoothing or interpolation is applied; corresponding AUC, final-success, and Hit90 values are reported in Table 1.

TD-error weighting and PER comparisons. SABER achieves higher mean AUC than TD-error critic-loss weighting in all seven cohorts. This is the closest allocation-level comparison because both methods preserve uniform replay, minibatch membership, Bellman targets, and critic-only application. The result shows that the matched TD-error weighting rule does not reproduce SABER's gains and distinguishes current-policy support from direct TD-error emphasis.

Across all seven cohorts, PER produces a mixed ordering. SABER has higher mean AUC on FPP DDPG, FPP TD3, and both DrawerClose backbones, whereas PER is higher on FPP SAC and both DrawerOpen backbones. DrawerClose TD3 also exposes a metric trade-off: SABER attains higher AUC (0.549 versus 0.524) and an earlier median Hit90 (80k versus 120k), while PER has marginally higher mean final success (0.600 versus 0.590). PER changes the replay distribution and applies importance correction, so it is a broader replay-selection intervention rather than a matched loss-allocation control. Neither replay prioritization nor critic-loss allocation therefore uniformly dominates across tasks, backbones, and metrics.

Taken together, these comparisons place SABER within

a broader family of HER-based learning improvements while isolating its distinct post-relabeling intervention point. SABER raises mean AUC over matched HER and TD-error critic-loss weighting in all seven cohorts and remains competitive with HGR and PER. Experience selection is therefore not the only consequential design decision in HER actor–critic training: critic-loss allocation within an already sampled and relabeled minibatch is a distinct and empirically consequential design axis.

4.3 Diagnostic Instrumentation

Diagnostic logging adds no gradient terms and does not alter replay sampling, training objectives, or the parameter-update order. For FPP, we record one full minibatch every 10k environment steps for seeds 1–5 of DDPG, TD3, and SAC, yielding 500 diagnostic batches per backbone and 1,500 in total. DrawerOpen and DrawerClose DDPG use separate ten-seed diagnostic cohorts. Snapshot-level quantities are first averaged within each seed, and seeds are treated as the independent analysis units.

Support activation. We first measure how often SABER's support condition activates during training. The support-positive rate is the fraction of logged samples for which $\psi_i = 1$. It is 0.367 ± 0.011, 0.219 ± 0.009, and 0.197 ± 0.013 for FPP DDPG, TD3, and SAC, respectively, and 0.429 ± 0.051 and 0.298 ± 0.013 for DrawerOpen and DrawerClose DDPG. Although activation varies across tasks and backbones, the indicator consistently identifies a nontrivial subset of each minibatch rather than degenerating to an almost-always or almost-never signal.

TD-error enrichment. We next relate support to the pre-update absolute Bellman residual $|\delta_i^{\mathrm{pre}}|$. For FPP DDPG, TD3, and SAC, Pearson correlations with the binary support indicator are weak at 0.116, 0.128, and 0.143. Nevertheless, support-positive rates within the top TD-error decile are enriched by $1.409\times$, $1.670\times$, and $1.828\times$ over their corresponding base rates. On DrawerOpen, mean pre-update absolute residuals are 0.291 for support-positive and 0.249 for support-negative samples, a $1.17\times$ ratio whose direction holds in all ten seeds. DrawerClose similarly combines a weak support–TD correlation of 0.111 with $1.568\times$ top-decile enrichment; its support-positive/negative residual means are $0.130/0.089$, or $1.45\times$, with the same direction in all ten seeds. Support therefore concentrates in high-error regions without reducing to a TD-error ranking.

One-update residual reduction. To test whether support-conditioned weights affect the intended critic update, we hold the Bellman targets fixed and measure $|\delta_i^{\mathrm{pre}}| - |\delta_i^{\mathrm{post}}|$ on the same minibatch after one critic step. Mean reductions for support-positive/negative samples are $0.0192/0.0080$, $0.0213/0.0065$, and $0.0223/0.0057$ for FPP DDPG, TD3, and SAC, corresponding to ratios of $2.39\times$, $3.30\times$, and $3.94\times$. The positive-sample reduction is larger in all 15 FPP backbone–seed runs. DrawerOpen and DrawerClose similarly give $0.00754/0.00359$ and $0.00628/0.00181$, or $2.10\times$ and $3.48\times$, respectively, with the same direction in all ten seeds of each cohort. These results verify that SABER translates its support signal into the intended redistribution of immediate critic fitting; the independent contribution of exact sample assignment is evaluated in the subsequent ablations.

Table 2: Support-signal diagnostics. Parentheses give the number of seeds. Support-positive rates are means $\pm$ sample standard deviations; ratios compare support-positive with support-negative means.

Cohort (n)	Support rate	Pre-TD ratio	1-step $\Delta\lvert\delta\rvert$
FPP DDPG (5)	0.367 ± 0.011	$1.42\times$	$2.39\times$
FPP TD3 (5)	0.219 ± 0.009	$1.62\times$	$3.30\times$
FPP SAC (5)	0.197 ± 0.013	$1.74\times$	$3.94\times$
DrawerOpen DDPG (10)	0.429 ± 0.051	$1.17\times$	$2.10\times$
DrawerClose DDPG (10)	0.298 ± 0.013	$1.45\times$	$3.48\times$

Phase-conditioned allocation. Finally, we examine where support activates along Drawer task progression using task-defined phases conditioned on the relabeled training goal. On DrawerOpen, support-positive rates are 0.737, 0.656, 0.641, and 0.326 for approach, contact, opening, and open, respectively; opening is aggregated over the nine seeds in which that phase is observed. Although open accounts for 73.8% of logged samples, it has the lowest support-positive rate. DrawerClose shows a complementary pattern: rates are 0.559, 0.719, 0.432, and 0.208 for approach, contact, closing, and closed, respectively, while closed accounts for 74.5% of samples and again has the lowest rate. Across both tasks, SABER therefore activates more frequently around approach and drawer interaction than in the dominant terminal phase. Full phase counts, batch fractions, and seed-level results are reported in the supplement.

4.4 Mechanism Ablations

The primary comparisons establish SABER's performance effect but do not identify whether it arises from the action used to query support, the pairing between that action and the sampled context, or the sample-wise placement of high critic weights. We therefore evaluate four controls with DDPG on FPP, DrawerOpen, and DrawerClose using paired seeds 1–5. Unless stated otherwise, each control retains Main SABER's task-specific training budget, HER and replay procedures, support hyperparameters, critic-only scope, UTD ratio, and evaluation protocol, without variant-specific retuning. Table 3 reports paired Main-minus-control AUC differences; positive values favor Main SABER.

Replay-Action Query. This control tests whether querying support with the current-policy action is more useful for learning than querying with the replay action. We replace a_i^π with a_i^r only in the support-membership query, while retaining the sample context, population thresholds, weight computation, Bellman regression at the replay action, and the unweighted actor objective. The sampled transition is not removed from its local replay pool, so its replay action can support itself when it appears within the bounded scan. Main SABER achieves higher mean AUC by $+0.011$, $+0.436$,

Table 3: Paired Main-minus-control normalized success AUC differences with 95% paired-bootstrap confidence intervals, using DDPG seeds 1–5. Positive values favor Main SABER.

Task	Replay-Action Query	Shuffled-Policy Query	Random-Support	Strict Count-Matched
FPP	$+0.011[-0.028, 0.059]$	$+0.043[-0.002, 0.085]$	$+0.056[0.014, 0.101]$	$+0.033[-0.006, 0.076]$
DrawerOpen	$+0.436[0.121, 0.729]$	$+0.105[-0.080, 0.335]$	$+0.047[-0.042, 0.136]$	$+0.149[-0.042, 0.442]$
DrawerClose	$+0.049[-0.006, 0.128]$	$+0.033[-0.063, 0.119]$	$+0.048[-0.004, 0.115]$	$+0.030[-0.033, 0.092]$

and $+0.049$ on FPP, DrawerOpen, and DrawerClose, respectively. The DrawerOpen confidence interval excludes zero, providing the clearest task-level evidence that using the current-policy action as the support query can materially affect learning.

Shuffled-Policy Query. This control examines whether SABER benefits from pairing each state–goal sample with the current-policy action computed for that sample. We preserve the complete multiset of current-policy actions within the minibatch but apply a fixed-point-free permutation so that every action is paired with a different sample's unchanged context. The shuffled action is used only to query support; the resulting weight remains attached to the original sample's replay-action residual. Main SABER achieves higher mean AUC by $+0.043$, $+0.105$, and $+0.033$ on FPP, DrawerOpen, and DrawerClose, respectively, although all three confidence intervals include zero. The consistent direction suggests that sample–query alignment contributes to SABER's performance. Replay-Action Query and Shuffled-Policy Query can also alter the support-positive count; they therefore measure the overall effects of query identity and pairing, including induced count changes, rather than isolating alignment under a matched count.

Random-Support. The query controls can change the number of samples receiving a high weight. Random-Support separates this count effect from the identities of the selected samples. At each control update, we apply Main SABER's indicator rule to that control run's own minibatch and replay state, obtaining a counterfactual Main mask M and positive count $k = \sum_i M_i$. We then draw exactly k minibatch positions uniformly without replacement and assign them the high raw weight. Random-Support therefore preserves Main's per-update positive count and raw and normalized weight distributions while removing sample-level context eligibility and action-bin membership alignment from the high-weight locations. Main SABER achieves higher mean AUC on all three tasks, and its $+0.056$ advantage on FPP has a confidence interval that excludes zero. This result indicates that SABER's gains cannot be explained solely by assigning larger weights to a matched number of arbitrary residuals. Because k is still obtained from Main's support computation, Random-Support is a count-matched control rather than a support-free random baseline.

Strict Context-Eligible Count-Matched. Random-Support can assign high weights to samples whose local replay contexts are not sufficiently populated. We therefore construct a stricter control that also matches context eligibility. At each update, we apply Main SABER's rule to the control run's own minibatch and replay state to obtain the counterfactual Main mask and its positive count k. We then form an eligible set E containing samples that satisfy Main SABER's local-context population conditions and draw exactly k samples uniformly without replacement from E. This control matches Main in positive count, raw and normalized weight distributions, and eligibility domain. By construction, its only intended remaining algorithmic difference from Main is whether the high-weight sample locations are selected by current-policy action-bin membership. Main SABER achieves higher mean AUC by $+0.033$, $+0.149$, and $+0.030$ on FPP, DrawerOpen, and DrawerClose, respectively, but all three confidence intervals include zero. The results are consistent with a contribution from membership-aligned weight assignment, but do not independently establish exact support membership as the unique causal mechanism.

References

Andrychowicz, M.; Wolski, F.; Ray, A.; Schneider, J.; Fong, R.; Welinder, P.; McGrew, B.; Tobin, J.; Abbeel, P.; and Zaremba, W. 2017. Hindsight Experience Replay. *Advances in Neural Information Processing Systems*.

Fang, M.; Zhou, T.; Du, Y.; Han, L.; and Zhang, Z. 2019. Curriculum-Guided Hindsight Experience Replay. In *Advances in Neural Information Processing Systems*, 12623–12634.

Fujimoto, S.; van Hoof, H.; and Meger, D. 2018. Addressing Function Approximation Error in Actor-Critic Methods. *Proceedings of the 35th International Conference on Machine Learning*.

Haarnoja, T.; Zhou, A.; Abbeel, P.; and Levine, S. 2018. Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor. *Proceedings of the 35th International Conference on Machine Learning*.

Lanka, S.; and Wu, T. 2018. ARCHER: Aggressive Rewards to Counter Bias in Hindsight Experience Replay. *arXiv preprint arXiv:1809.02070*.

Lillicrap, T. P.; Hunt, J. J.; Pritzel, A.; Heess, N.; Erez, T.; Tassa, Y.; Silver, D.; and Wierstra, D. 2015. Continuous Control with Deep Reinforcement Learning. *arXiv preprint arXiv:1509.02971*.

Luu, T. M.; and Yoo, C. D. 2021. Hindsight Goal Ranking on Replay Buffer for Sparse Reward Environment. *IEEE Access*.

Novati, G.; and Koumoutsakos, P. 2019. Remember and Forget for Experience Replay. In *Proceedings of the 36th*

International Conference on Machine Learning, volume 97 of *Proceedings of Machine Learning Research*, 4851–4860.

Ren, Z.; Dong, K.; Zhou, Y.; Liu, Q.; and Peng, J. 2019. Exploration via Hindsight Goal Generation. In *Advances in Neural Information Processing Systems*.

Schaul, T.; Quan, J.; Antonoglou, I.; and Silver, D. 2015. Prioritized Experience Replay. *arXiv preprint arXiv:1511.05952*.

Schramm, L.; Deng, Y.; Granados, E.; and Boularias, A. 2023. USHER: Unbiased Sampling for Hindsight Experience Replay. In *Proceedings of the 6th Conference on Robot Learning*, volume 205 of *Proceedings of Machine Learning Research*, 2073–2082.

Yang, R.; Lyu, J.; Yang, Y.; Yan, J.; Luo, F.; Luo, D.; Li, L.; and Li, X. 2021. Bias-Reduced Multi-Step Hindsight Experience Replay for Efficient Multi-Goal Reinforcement Learning. *arXiv preprint arXiv:2102.12962*.

Zhao, R.; and Tresp, V. 2018. Energy-Based Hindsight Experience Prioritization. In *Proceedings of the Conference on Robot Learning*, 113–122.